NUMERICAL IITAKA DIMENSIONS DEFINED BY SECTIONS AND BY VOLUMES COINCIDE

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Abstract. We prove that two different definitions of numerical Iitaka dimensions, one defined by the growth of sections and the other by the growth of volumes, coincide. This answers a question of Jiang and Wang.

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1. Introduction

We work over the field of complex numbers $\mathbb{C}$.

Let X be a normal projective variety of dimension d and D a pseudo-effective $\mathbb{R}$-Cartier $\mathbb{R}$-divisor on X. In all limits as $m \to +\infty$ below, $m \in \mathbb{N}^+$. When X is smooth, Nakayama [Nak04, Chapter V, Definition 2.5] defined the numerical dimension

$$\kappa_\sigma(D) := \max\left\{\ell \in \mathbb{N} \;\middle|\; \limsup_{m\to+\infty} \frac{h^0(X, \lfloor mD \rfloor + A)}{m^\ell} > 0 \text{ for some divisor } A \text{ on } X\right\}.$$

Lehmann [Leh13] considered the volume type invariant

$$\nu_{\mathrm{vol}}(D) := \max\left\{\ell \in \mathbb{N} \;\middle|\; \mathrm{vol}_X(D + tA) > Ct^{d-\ell} \text{ for some real number } C > 0 \text{ and any } t > 0\right\},$$

where A is a fixed sufficiently ample divisor on X and vol_X is the volume (see Definition 2.1). For smooth X, [Leh13, Theorem 1.1] and [Eck16, Theorem 0.2] state in particular that $\nu_{\mathrm{vol}}(D) = \kappa_\sigma(D)$. However, [Les22, Remark 4] points out that both papers have gaps: [Leh13, Proposition 5.3] fails in general, and the proof of [Eck16, Proposition 3.4] fails.

When X is smooth, Lesieutre [Les22, Definition 1] introduced the real-valued invariant

$$\kappa^{\mathbb{R}}_{\sigma,\sup}(X, D) := \sup\left\{\ell \in \mathbb{R} \;\middle|\; \limsup_{m\to+\infty} \frac{h^0(X, \lfloor mD \rfloor + A)}{m^\ell} > 0 \text{ for some ample divisor } A \text{ on } X\right\}$$

(see Definition 2.1). On a smooth Calabi–Yau threefold X of Picard number 2, Lesieutre [Les22, Section 3] gave a pseudo-effective $\mathbb{R}$-divisor D_+ such that

$$\kappa^{\mathbb{R}}_{\sigma,\sup}(X, D_+) = 3/2 \quad \text{and} \quad \kappa_\sigma(D_+) = 1.$$

Moreover, [Les22, Remark 8] asks whether $\kappa_\sigma(D) = \nu_{\mathrm{vol}}(D)$ in general.

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Choi and Park [CP21, Section 1] introduced real-valued invariants of both types on a $\mathbb{Q}$-factorial X. Let A be a fixed sufficiently positive ample $\mathbb{Z}$-divisor on X. For $\diamond \in \{\sup, \inf\}$ and $\lim^{\diamond}$ as in Definition 2.1, they defined the section type invariant and the volume type invariant

$$\widehat{\kappa}^{\mathbb{R}}_{\sigma,\diamond}(D) := \sup\left\{\ell \in \mathbb{R}_{\geq 0} \;\middle|\; \lim_{m\to+\infty}^{\diamond} \frac{h^0(X, \lfloor mD \rfloor + A)}{m^\ell} > 0\right\},$$
$$\widehat{\kappa}^{\mathbb{R}}_{\mathrm{vol},\diamond}(D) := \sup\left\{\ell \in \mathbb{R}_{\geq 0} \;\middle|\; \lim_{t\to 0^+}^{\diamond} \frac{\mathrm{vol}_X(D + tA)}{t^{d-\ell}} > 0\right\}.$$

Compared with Definition 2.1:

(1) A is fixed in $\widehat{\kappa}^{\mathbb{R}}_{\sigma,\diamond}(D)$;
(2) $\ell \in \mathbb{R}_{\geq 0}$;
(3) $\widehat{\kappa}^{\mathbb{R}}_{\mathrm{vol},\diamond}(D)$ uses $\mathrm{vol}_X(D + tA)$ with $t \to 0^+$ instead of $\mathrm{vol}_X(mD + A)$ with $m \to +\infty$.

[CP21, Lemma 3.3] shows that, if X is $\mathbb{Q}$-factorial, D is pseudo-effective, and A is a fixed sufficiently positive ample $\mathbb{Z}$-divisor on X, then

$$\mathrm{vol}_X(mD + 2A) \geq h^0(X, \lfloor mD \rfloor + A) \quad \text{for any } m \in \mathbb{N}^+.$$

Using [CP21, Lemma 3.3], [CP21, Proposition 3.1] proves that

$$\widehat{\kappa}^{\mathbb{R}}_{\sigma,\diamond}(D) \leq \widehat{\kappa}^{\mathbb{R}}_{\mathrm{vol},\diamond}(D) \quad \text{for } \diamond \in \{\sup, \inf\},$$

and [CP21, Question 3.4] asks whether equality holds for $\diamond \in \{\sup, \inf\}$.

Jiang and Wang [JW24, Definition 1.1] extended $\kappa^{\mathbb{R}}_{\sigma,\sup}(X, D)$ to any normal projective variety X, and defined $\kappa^{\mathbb{R}}_{\sigma,\inf}(X, D)$ by replacing lim sup with lim inf. Fixing an ample Cartier divisor A on X, they also defined the volume type invariant $\kappa^{\mathbb{R}}_{\mathrm{vol},\diamond}(X, D)$ (see Definition 2.1), which does not depend on A [JW24, Lemma 2.2(1)]. For $\diamond = \inf$, [JW24, Definition 2.1] defines $\kappa^{\mathbb{R}}_{\mathrm{vol},\inf}(X, D)$ as the supremum of $\ell \in \mathbb{R}$ such that $\mathrm{vol}_X(D + tA) > Ct^{d-\ell}$ for some real number $C > 0$ and any $t > 0$, and notes that, by [CP21, Lemma 3.2], the condition on $\mathrm{vol}_X(D + tA)$ is equivalent to $\liminf_{m\to+\infty} \mathrm{vol}_X(mD + A)/m^\ell > 0$. For $\diamond = \sup$, [JW24, Definition 2.1] uses the condition $\limsup_{m\to+\infty} \mathrm{vol}_X(mD + A)/m^\ell > 0$. Definition 2.1 uses the conditions with $m \to +\infty$ for both $\diamond$. By [JW24, Lemma 2.2(3)], for any $k \in \mathbb{N}^+$,

$$\kappa^{\mathbb{R}}_{\sigma,\diamond}(X, kD) = \kappa^{\mathbb{R}}_{\sigma,\diamond}(X, D).$$

[JW24, Remark 1.8(2)] states the inequalities

$$\kappa^{\mathbb{R}}_{\sigma,\diamond}(X, D) \leq \kappa^{\mathbb{R}}_{\mathrm{vol},\diamond}(X, D) \quad \text{for } \diamond \in \{\sup, \inf\}$$

in general, with reference to [CP21, Proposition 3.1], which compares $\widehat{\kappa}^{\mathbb{R}}_{\sigma,\diamond}(D)$ with $\widehat{\kappa}^{\mathbb{R}}_{\mathrm{vol},\diamond}(D)$ on a $\mathbb{Q}$-factorial X. We prove the inequalities between $\kappa^{\mathbb{R}}_{\sigma,\diamond}(X, D)$ and $\kappa^{\mathbb{R}}_{\mathrm{vol},\diamond}(X, D)$ in Section 5. Based on [Les22, Remark 8] and [CP21, Question 3.4], Jiang and Wang [JW24, Question 1.5] asked whether the equalities

$$\kappa^{\mathbb{R}}_{\sigma,\sup}(X, D) = \kappa^{\mathbb{R}}_{\mathrm{vol},\sup}(X, D)$$

and

$$\kappa^{\mathbb{R}}_{\sigma,\inf}(X, D) = \kappa^{\mathbb{R}}_{\mathrm{vol},\inf}(X, D)$$

hold for any normal projective variety X and any pseudo-effective $\mathbb{R}$-Cartier $\mathbb{R}$-divisor D on X.

Filip, Lesieutre and Tosatti [FLT26, Theorem 1.3.1] showed that the limsup type and the liminf type growth exponents can differ: for any integer $d \geq 3$, they constructed a smooth Calabi–Yau variety X of dimension d such that, for any real number $\delta \in [1, d/2]$, there exists a pseudo-effective $\mathbb{R}$-divisor D on X with

$$\limsup_{m\to+\infty} \frac{\log h^0(X, \lfloor mD \rfloor + A)}{\log m} = d - \delta \quad \text{and} \quad \liminf_{m\to+\infty} \frac{\log h^0(X, \lfloor mD \rfloor + A)}{\log m} = \frac{d}{2}$$

for any sufficiently ample divisor A on X. In particular,

$$\kappa^{\mathbb{R}}_{\sigma,\sup}(X,D) = d-\delta > \frac{d}{2} = \kappa^{\mathbb{R}}_{\sigma,\inf}(X,D) \quad \text{if } \delta < d/2.$$

Moreover, for $\delta = 1$ and $d \geq 5$, [FLT26, Corollary 1.5.4] shows that $\kappa_\sigma(D) \geq d-2 > \lfloor d/2 \rfloor = \nu_{\mathrm{vol}}(D)$, which answers the question of [Les22, Remark 8] negatively. The inequality $\kappa_\sigma(D) > \nu_{\mathrm{vol}}(D)$ compares the lim sup type invariant $\kappa_\sigma(D)$ with the lim inf type invariant $\nu_{\mathrm{vol}}(D)$, and does not contradict Theorem 1.1.

The equality of the section type and the volume type invariants was known in the following cases:

(1) When X is $\mathbb{Q}$-factorial, D is nef, and the invariants are $\widehat{\kappa}^{\mathbb{R}}_{\sigma,\diamond}(D)$ and $\widehat{\kappa}^{\mathbb{R}}_{\mathrm{vol},\diamond}(D)$ [CP21].
(2) When X is a Calabi–Yau variety of finite birational index in the sense of [JW24, Definitions 1.2 and 1.6] and the class of D is in the closed movable cone of X, both equalities of [JW24, Question 1.5] hold [JW24].

In general, [JW24, Question 1.5] was open. The goal of this paper is to prove both equalities of [JW24, Question 1.5].

Theorem 1.1. *Let X be a normal projective variety and D a pseudo-effective $\mathbb{R}$-Cartier $\mathbb{R}$-divisor on X. Then*

$$\kappa^{\mathbb{R}}_{\sigma,\sup}(X,D) = \kappa^{\mathbb{R}}_{\mathrm{vol},\sup}(X,D) \quad \textit{and} \quad \kappa^{\mathbb{R}}_{\sigma,\inf}(X,D) = \kappa^{\mathbb{R}}_{\mathrm{vol},\inf}(X,D).$$

Sketch of the proof. Let $f : Y \to X$ be a resolution and $\diamond \in \{\sup, \inf\}$. By Proposition 3.1,

$$\kappa^{\mathbb{R}}_{\sigma,\diamond}(Y, f^*D) = \kappa^{\mathbb{R}}_{\sigma,\diamond}(X,D) \quad \text{and} \quad \kappa^{\mathbb{R}}_{\mathrm{vol},\diamond}(Y, f^*D) = \kappa^{\mathbb{R}}_{\mathrm{vol},\diamond}(X,D).$$

Thus we may assume that X is smooth. By [CP21, Lemma 3.3],

$$\kappa^{\mathbb{R}}_{\sigma,\diamond}(X,D) \leq \kappa^{\mathbb{R}}_{\mathrm{vol},\diamond}(X,D).$$

Conversely, by Theorem 4.1, for any ample Cartier divisor A on X, there exists an ample Cartier divisor A' on X such that

$$h^0(X, \lfloor (d+1)mD \rfloor + A') \geq \mathrm{vol}_X(mD + A) \quad \text{for any } m \in \mathbb{N}^+.$$

Thus

$$\kappa^{\mathbb{R}}_{\mathrm{vol},\diamond}(X,D) \leq \kappa^{\mathbb{R}}_{\sigma,\diamond}(X,(d+1)D) = \kappa^{\mathbb{R}}_{\sigma,\diamond}(X,D),$$

where the equality is [JW24, Lemma 2.2(3)].

For a very ample divisor H on X and a divisor $G \geq 0$ on X with $G \sim K_X + (d+1)H$, Lemmas 4.2 and 4.3 imply (see the proof of Theorem 4.1)

$$h^0(X, K_X + (d+1)(L+G)) \geq \mathrm{vol}_X(L) \quad \text{for any big divisor } L \text{ on } X. \tag{1.1}$$

Theorem 4.1 follows from (1.1) for L equal to $\lfloor mD \rfloor$ plus a fixed divisor positive enough that $L \geq mD + A$ for any $m \in \mathbb{N}^+$ and $A' := K_X + (d+1)(L+G) - (d+1)\lfloor mD \rfloor$ is ample: since $L - \lfloor mD \rfloor$ and G do not depend on m, A' does not depend on m, and $\lfloor (d+1)x \rfloor \geq (d+1)\lfloor x \rfloor$ for any real number x. Fujita's approximation theorem [Fuj94], which approximates $\mathrm{vol}_X(L)$ up to a given $\epsilon > 0$ by self-intersection numbers of ample $\mathbb{Q}$-divisors on higher models, is proved in [DEL00, Theorem 3.2] with [DEL00, Theorem 1.8(iii)] for $nL - G$ with $n \gg 0$. This proof obtains a base point free divisor M_n on a log resolution with $(M_n^d)/n^d \geq \mathrm{vol}_X(L) - \epsilon$. The proof of Lemma 4.2 uses [DEL00, Theorem 1.8(iii)] for L itself and obtains the base point free divisor M of the lemma with $(M^d) \geq \mathrm{vol}_X(L)$, and G goes into A'.

Case 2 is [JW24, Theorem 1.7], proved with [JW24, Proposition 3.4] in the role of Theorem 4.1: for X and D as in case 2 and some ample Cartier divisor A on X,

$$h^0(X, \lfloor NmD \rfloor + 2NA) > CN^d \,\mathrm{vol}_X(mD + A) \quad \text{for any } m \in \mathbb{N}^+,$$

where the positive integer N and the real number $C > 0$ do not depend on m. Theorem 4.1 is such an inequality, with $\geq$ in place of $>$, $N = d+1$, the twist A' in place of $2NA$, and 1 in place of CN^d, for any smooth projective variety X and any $\mathbb{R}$-divisor D on X.

Structure of the paper. In Section 2 we recall the volume and the four real numerical Iitaka dimensions of [JW24]. In Section 3 we show that the four invariants do not change under pullback to a higher model. In Section 4 we bound the dimensions of spaces of sections from below by volumes on a smooth projective variety. In Section 5 we prove the main theorem.

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2. Preliminaries

Definition 2.1 ([JW24, Definitions 1.1 and 2.1])**.** Let X be a normal projective variety of dimension d. For a Weil divisor W on X, let $h^0(X,W) := \dim H^0(X,\mathcal{O}_X(W))$. In the limits below, m and n are positive integers, and $\sup\emptyset := -\infty$.

(1) For an $\mathbb{R}$-Cartier $\mathbb{R}$-divisor L on X, the *volume* of L is

$$\mathrm{vol}_X(L) := \limsup_{n\to+\infty} \frac{h^0(X,\lfloor nL\rfloor)}{n^d/d!}.$$

(2) Let D be a pseudo-effective $\mathbb{R}$-Cartier $\mathbb{R}$-divisor on X and $\diamond \in \{\sup,\inf\}$. Let $\lim^\diamond$ denote $\limsup$ if $\diamond = \sup$ and $\liminf$ if $\diamond = \inf$. We define

$$\kappa^{\mathbb{R}}_{\sigma,\diamond}(X,D) := \sup\Big\{\ell\in\mathbb{R} \;\Big|\; \lim^{\diamond}_{m\to+\infty} \frac{h^0(X,\lfloor mD\rfloor + A)}{m^\ell} > 0 \text{ for some ample Cartier divisor } A \text{ on } X\Big\}.$$

For an ample Cartier divisor A on X, we define

$$\kappa^{\mathbb{R}}_{\mathrm{vol},\diamond}(X,D) := \sup\left\{\ell\in\mathbb{R} \;\middle|\; \lim^{\diamond}_{m\to+\infty} \frac{\mathrm{vol}_X(mD+A)}{m^\ell} > 0\right\},$$

which does not depend on the choice of A by [JW24, Lemma 2.2(1)]. We call $\kappa^{\mathbb{R}}_{\sigma,\diamond}(X,D)$ and $\kappa^{\mathbb{R}}_{\mathrm{vol},\diamond}(X,D)$ the *numerical Iitaka dimensions* of D.

3. Reduction to smooth varieties

Proposition 3.1. *Let $f: Y\to X$ be a birational morphism between normal projective varieties and D a pseudo-effective $\mathbb{R}$-Cartier $\mathbb{R}$-divisor on X. Then f^*D is pseudo-effective and, for any $\diamond\in\{\sup,\inf\}$,*

$$\kappa^{\mathbb{R}}_{\sigma,\diamond}(Y,f^*D) = \kappa^{\mathbb{R}}_{\sigma,\diamond}(X,D) \quad \text{and} \quad \kappa^{\mathbb{R}}_{\mathrm{vol},\diamond}(Y,f^*D) = \kappa^{\mathbb{R}}_{\mathrm{vol},\diamond}(X,D).$$

Proof. By [Nak04, Lemma II.2.11],

$$f_*\mathcal{O}_Y(\lfloor f^*L\rfloor) = \mathcal{O}_X(\lfloor L\rfloor),$$

so $h^0(Y,\lfloor f^*L\rfloor) = h^0(X,\lfloor L\rfloor)$ for any $\mathbb{R}$-Cartier $\mathbb{R}$-divisor L on X.

Let A_X be an ample Cartier divisor on X and A_Y an ample Cartier divisor on Y. Since A_Y is ample and f^*A_X is big, by Kodaira's lemma there exist positive integers a and b such that

$$|aA_Y - f^*A_X| \neq \emptyset \quad \text{and} \quad |bf^*A_X - A_Y| \neq \emptyset.$$

Let m be a positive integer. Taking $L = mD + A_X$, and $L = n(mD+A_X)$ for any positive integer n, we obtain

$$h^0(Y,\lfloor mf^*D\rfloor + f^*A_X) = h^0(X,\lfloor mD\rfloor + A_X) \tag{3.1}$$

and

$$\mathrm{vol}_Y(mf^*D + f^*A_X) = \mathrm{vol}_X(mD+A_X). \tag{3.2}$$

Since $|aA_Y - f^*A_X| \neq \emptyset$ and $|bf^*A_X - A_Y| \neq \emptyset$, (3.1) and (3.2) for A_X and for bA_X give

$$h^0(X, \lfloor mD \rfloor + A_X) = h^0(Y, \lfloor mf^*D \rfloor + f^*A_X) \leq h^0(Y, \lfloor mf^*D \rfloor + aA_Y), \tag{3.3}$$

$$h^0(Y, \lfloor mf^*D \rfloor + A_Y) \leq h^0(Y, \lfloor mf^*D \rfloor + bf^*A_X) = h^0(X, \lfloor mD \rfloor + bA_X), \tag{3.4}$$

$$\mathrm{vol}_X(mD + A_X) = \mathrm{vol}_Y(mf^*D + f^*A_X) \leq \mathrm{vol}_Y(mf^*D + aA_Y), \tag{3.5}$$

$$\mathrm{vol}_Y(mf^*D + A_Y) \leq \mathrm{vol}_Y(mf^*D + bf^*A_X) = \mathrm{vol}_X(mD + bA_X). \tag{3.6}$$

Since A_X and A_Y are arbitrary and $\kappa^{\mathbb{R}}_{\mathrm{vol},\diamond}$ does not depend on the ample Cartier divisor (Definition 2.1), (3.3)–(3.6) imply both equalities for any $\diamond \in \{\sup, \inf\}$. The proposition follows. □

4. Bounding volumes by sections on smooth varieties

Theorem 4.1. *Let X be a smooth projective variety of dimension d, D an $\mathbb{R}$-divisor on X, and A an ample Cartier divisor on X. Then there exists an ample Cartier divisor A' on X such that*

$$h^0(X, \lfloor (d+1)mD \rfloor + A') \geq \mathrm{vol}_X(mD + A)$$

for any positive integer m.

Lemma 4.2 (cf. [DEL00, Section 3])**.** *Let X be a smooth projective variety of dimension d, H a very ample divisor on X, and $G \geq 0$ a divisor on X such that $G \sim K_X + (d+1)H$. Let L be a big divisor on X. Then there exist a birational morphism $f : Y \to X$ from a smooth projective variety Y and a base point free divisor M on Y such that $f^*(L+G) - M \geq 0$ and*

$$(M^d) \geq \mathrm{vol}_X(L).$$

Proof. Let $\mathcal{J} := \mathcal{J}(\|L\|)$ be the asymptotic multiplier ideal of L [DEL00, Definition 1.7]. Since H is very ample and $L + G \sim K_X + L + (d+1)H$, by [DEL00, Theorem 1.8(iii)],

$$\mathcal{O}_X(L+G) \otimes \mathcal{J} \text{ is globally generated.}$$

Let $f : Y \to X$ be a log resolution of $\mathcal{J}$. Then Y is a smooth projective variety, and $\mathcal{J} \cdot \mathcal{O}_Y = \mathcal{O}_Y(-F)$ for some divisor $F \geq 0$ on Y. Let

$$M := f^*(L+G) - F.$$

Pulling back to Y the global sections of $\mathcal{O}_X(L+G) \otimes \mathcal{J}$, we see that

$$\mathcal{O}_Y(M) = f^*\mathcal{O}_X(L+G) \otimes \mathcal{O}_Y(-F)$$

is globally generated, that is, M is base point free.

Let n be a positive integer. Since nL is big, by [DEL00, Theorem 1.8(i)] for nL and by [DEL00, Variant 2.4],

$$H^0(X, \mathcal{O}_X(nL)) = H^0(X, \mathcal{O}_X(nL) \otimes \mathcal{J}(\|nL\|)) \subset H^0(X, \mathcal{O}_X(nL) \otimes \mathcal{J}^n).$$

Since f is birational and $\mathcal{J}^n \cdot \mathcal{O}_Y = \mathcal{O}_Y(-nF)$, pulling back sections to Y is injective on the last space and takes values in $H^0(Y, \mathcal{O}_Y(nf^*L - nF))$. Multiplying by the pullback of a section of $\mathcal{O}_X(nG)$ defining $nG \geq 0$, we obtain injections

$$\begin{aligned} H^0(X, \mathcal{O}_X(nL)) &\hookrightarrow H^0(Y, \mathcal{O}_Y(nf^*L - nF)) \\ &\hookrightarrow H^0(Y, \mathcal{O}_Y(nf^*L + nf^*G - nF)) = H^0(Y, \mathcal{O}_Y(nM)). \end{aligned}$$

Thus $h^0(Y, nM) \geq h^0(X, nL)$ for any positive integer n, so $\mathrm{vol}_Y(M) \geq \mathrm{vol}_X(L)$. Since M is nef, $\mathrm{vol}_Y(M) = (M^d)$ by [Laz04, Corollary 1.4.41]. Thus

$$(M^d) \geq \mathrm{vol}_X(L).$$

The lemma follows. □

Lemma 4.3. *Let X be a smooth projective variety of dimension d and M a big base point free divisor on X. Then*

$$h^0(X, K_X + (d+1)M) \geq (M^d).$$

Proof. We prove the lemma by induction on d. If $d = 0$, then X is a point and both sides equal 1.

Assume that $d = 1$. By Riemann–Roch, $h^0(X, K_X + 2M) = g(X) - 1 + 2\deg M \geq \deg M$.

Assume that $d \geq 2$ and that the lemma holds in dimension $d-1$. Let $S \in |M|$ be a general member. By Bertini and since $H^1(X, \mathcal{O}_X(-M)) = 0$ by Kawamata–Viehweg vanishing, S is a smooth projective variety of dimension $d-1$.

Let $M_S := M|_S$. Then M_S is base point free and, since M is nef and big,

$$(M_S^{d-1}) = (M^d) > 0.$$

Thus M_S is big. By the induction hypothesis,

$$h^0(S, K_S + dM_S) \geq (M_S^{d-1}) = (M^d).$$

By adjunction, $(K_X + (d+1)M)|_S \sim K_S + dM_S$, so there exists an exact sequence

$$0 \to \mathcal{O}_X(K_X + dM) \to \mathcal{O}_X(K_X + (d+1)M) \to \mathcal{O}_S(K_S + dM_S) \to 0.$$

Since dM is nef and big, $H^1(X, \mathcal{O}_X(K_X + dM)) = 0$ by Kawamata–Viehweg vanishing. Thus the restriction map

$$H^0(X, \mathcal{O}_X(K_X + (d+1)M)) \to H^0(S, \mathcal{O}_S(K_S + dM_S))$$

is surjective, and

$$h^0(X, K_X + (d+1)M) \geq h^0(S, K_S + dM_S) \geq (M^d).$$

The lemma follows. □

Proof of Theorem 4.1. Let $H \geq 0$ be a very ample divisor on X and $G \geq 0$ a divisor on X such that $G \sim K_X + (d+1)H$. Such H and G exist: take $H := kH_0$ for a very ample divisor $H_0 \geq 0$ on X and a sufficiently large positive integer k. Let q be a positive integer such that

$$A' := K_X + (d+1)(A + \operatorname{Supp} D + qH + G)$$

is ample.

Let m be a positive integer and let

$$L_m := \lfloor mD \rfloor + A + \operatorname{Supp} D + qH.$$

The coefficients of $mD - \lfloor mD \rfloor$ belong to $[0, 1)$ and $\operatorname{Supp}(mD - \lfloor mD \rfloor) \subset \operatorname{Supp} D$, so

$$L_m - (mD + A) = \big(\operatorname{Supp} D - (mD - \lfloor mD \rfloor)\big) + qH \geq 0.$$

Thus $\lfloor nL_m \rfloor \geq \lfloor n(mD + A) \rfloor$ for any positive integer n, and hence

$$\operatorname{vol}_X(L_m) \geq \operatorname{vol}_X(mD + A).$$

If $\operatorname{vol}_X(mD + A) = 0$, there is nothing to prove. Thus we may assume that $\operatorname{vol}_X(mD + A) > 0$. Then $\operatorname{vol}_X(L_m) > 0$, so L_m is big.

By Lemma 4.2 for $L = L_m$, there exist a birational morphism $f : Y \to X$ from a smooth projective variety Y and a base point free divisor M on Y such that $f^*(L_m + G) - M \geq 0$ and $(M^d) \geq \operatorname{vol}_X(L_m)$. Since M is nef and $(M^d) \geq \operatorname{vol}_X(L_m) > 0$, M is big. By Lemma 4.3,

$$h^0(Y, K_Y + (d+1)M) \geq (M^d) \geq \operatorname{vol}_X(L_m).$$

Multiplying by a section defining $(d+1)(f^*(L_m + G) - M) \geq 0$, we obtain an injection

$$H^0(Y, \mathcal{O}_Y(K_Y + (d+1)M)) \hookrightarrow H^0(Y, \mathcal{O}_Y(K_Y + (d+1)f^*(L_m + G))).$$

Since

$$K_Y + (d+1)f^*(L_m + G) = f^*\big(K_X + (d+1)(L_m + G)\big) + (K_Y - f^*K_X)$$

and $K_Y - f^*K_X \geq 0$ is f-exceptional,

$$h^0(Y, K_Y + (d+1)f^*(L_m + G)) = h^0(X, K_X + (d+1)(L_m + G)).$$

By the definitions of L_m and A',

$$K_X + (d+1)(L_m + G) = (d+1)\lfloor mD \rfloor + A'.$$

Since $\lfloor (d+1)x \rfloor \geq (d+1)\lfloor x \rfloor$ for any real number x, $\lfloor (d+1)mD \rfloor - (d+1)\lfloor mD \rfloor \geq 0$. Thus

$$\begin{aligned} h^0(X, \lfloor (d+1)mD \rfloor + A') &\geq h^0(X, (d+1)\lfloor mD \rfloor + A') = h^0(X, K_X + (d+1)(L_m + G)) \\ &= h^0(Y, K_Y + (d+1)f^*(L_m + G)) \geq h^0(Y, K_Y + (d+1)M) \\ &\geq \mathrm{vol}_X(L_m) \geq \mathrm{vol}_X(mD + A). \end{aligned}$$

The theorem follows. □

5. Proof of the main theorem

Proof of Theorem 1.1. By Proposition 3.1, possibly replacing X with a resolution $f : Y \to X$ and D with f^*D, we may assume that X is smooth. Let $d := \dim X$ and $\diamond \in \{\sup, \inf\}$.

Let A be an ample Cartier divisor on X. By Theorem 4.1, there exists an ample Cartier divisor A' on X such that

$$h^0(X, \lfloor m(d+1)D \rfloor + A') \geq \mathrm{vol}_X(mD + A)$$

for any positive integer m. Let ℓ be a real number such that

$$\operatorname*{\diamond\lim}_{m \to +\infty} \mathrm{vol}_X(mD + A)/m^\ell > 0.$$

Then

$$\operatorname*{\diamond\lim}_{m \to +\infty} h^0(X, \lfloor m(d+1)D \rfloor + A')/m^\ell > 0,$$

so

$$\ell \leq \kappa^{\mathbb{R}}_{\sigma,\diamond}(X, (d+1)D) = \kappa^{\mathbb{R}}_{\sigma,\diamond}(X, D),$$

where the equality follows from [JW24, Lemma 2.2(3)]. By Definition 2.1, $\kappa^{\mathbb{R}}_{\mathrm{vol},\diamond}(X, D)$ is the supremum of such ℓ, so

$$\kappa^{\mathbb{R}}_{\mathrm{vol},\diamond}(X, D) \leq \kappa^{\mathbb{R}}_{\sigma,\diamond}(X, D).$$

Let ℓ be a real number and A an ample Cartier divisor on X such that

$$\operatorname*{\diamond\lim}_{m \to +\infty} h^0(X, \lfloor mD \rfloor + A)/m^\ell > 0.$$

Let c be a sufficiently large integer. By [CP21, Lemma 3.3] for the ample Cartier divisor cA, and since $(c-1)A$ is linearly equivalent to an effective divisor,

$$\mathrm{vol}_X(mD + 2cA) \geq h^0(X, \lfloor mD \rfloor + cA) \geq h^0(X, \lfloor mD \rfloor + A)$$

for any positive integer m. Thus

$$\operatorname*{\diamond\lim}_{m \to +\infty} \mathrm{vol}_X(mD + 2cA)/m^\ell > 0.$$

By Definition 2.1 with $2cA$ in place of A, $\ell \leq \kappa^{\mathbb{R}}_{\mathrm{vol},\diamond}(X, D)$. Thus

$$\kappa^{\mathbb{R}}_{\sigma,\diamond}(X, D) \leq \kappa^{\mathbb{R}}_{\mathrm{vol},\diamond}(X, D).$$

The theorem follows. □

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